

Differential Diagnosis of COVID-19, Pneumonia, and Normal Findings: An Approach Based on Deep Learning and Uncertainty Management

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Abstract

The aim of this study is to develop and validate an artificial intelligence system for analysing chest X-rays. This system should be capable of distinguishing between different lung diseases and explicitly managing diagnostic uncertainty. The goal is to reduce the risk of false positives and false negatives. Four deep learning architectures (ResNet-50, ConvNeXt-Small, EfficientNet-B3 and DenseNet-121) were used and compared for classifying X-rays into three categories: Covid-19, normal and viral pneumonia. The impact of introducing a dynamic threshold for identifying ambiguous cases was also evaluated. These cases were assigned to a rejection class called UNKNOWN. Finally, we analyse the latent space topologically using the t-SNE algorithm to highlight the nature of the rejected images. We also visually investigate the source of decision uncertainty through activation maps (Grad-CAM), comparing cases with high and low predictive confidence.

Keywords: Deep Learning, COVID-19, X-rays, Uncertainty, Grad-CAM, t-SNE

1. Introduction

The overarching goal of this project is to develop and validate a computer vision framework for chest imaging diagnostics. The project aims to overcome the limitations of 'closed' classification systems by creating a model that can accurately distinguish between different lung conditions and natively integrates a safety mechanism for managing uncertainty — a critical feature in real-world healthcare systems.

The primary objective of the study is to determine the effectiveness of four convolutional neural networks (CNNs) in differentiating between cases of SARS-CoV-2 infection, viral pneumonia, and normal findings. In addition to validating the most effective model, the research aims to verify whether implementing a dynamic threshold adjusted via a sensitivity parameter (K) enables the correct classification of ambiguous cases as 'UNKNOWN', thereby reducing the rate of forced diagnostic errors.

A secondary objective involves XAI (Explainable AI), which aims to understand how the model

makes decisions and why it makes mistakes. This was achieved by performing a topological analysis of the model's latent space using the t-SNE algorithm. This demonstrated that rejected images are mathematically located at the boundaries between clusters associated with different pathologies. Interpretability was validated by using Grad-CAM. (Gradient-weighted Class Activation Mapping) technique, which maps the network's areas of focus by comparing the visual patterns of high-confidence cases with those of ambiguous cases.

2. Methodology

This section provides a detailed description of the technical procedures used for preparing the data, conducting an exploratory analysis of radiographic features, training deep learning models and implementing the uncertainty management system. All images used in this study were taken from the 'Covid-19 Radiography Database', which is available to download for free from Kaggle¹.

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¹For more information about the dataset, see <https://www.kaggle.com/datasets/tawsifurrahman/covid19-radiography-database>

2.1. Data Pipeline

Examining the original dataset revealed significant imbalances among the diagnostic categories: 10,192 samples were in the 'Normal' class, 3,616 were in the 'Covid' class and only 1,345 were in the 'Pneumonia' class. To prevent predictive biases favouring the majority class from developing, a rigorous undersampling procedure was applied to reduce the more numerous categories, resulting in a perfectly balanced dataset of 1,345 images for each of the three classes. Subsequently, the data were stratified according to the 'gold standard' (60%-20%-20%), and divided into training (807 images per class), validation (269 images per class) and test (269 images per class) sets. This ensured that the proportion of each class remained balanced in each subset, providing a reliable basis for evaluation.

2.2. Exploratory Data Analysis

Prior to the modelling phase, an exploratory analysis was conducted to extract and quantify the physical patterns inherent in the images. Figure (1) shows the analysis of the pixel intensity distribution. The x-axis represents the pixel intensity value, ranging from 0 (black) to 255 (pure white), while the y-axis represents the count (or frequency) of pixels with a specific intensity within the analysed dataset for that particular class. Compared to the other two classes, the distribution of pixels for the 'Covid' class is noticeably shifted to the right and has a higher average pixel value; this means that X-rays of lungs affected by 'Covid' are brighter.



Figure 1: Pixels analysis for each class

2.3. Deep Learning Architecture and Training Pipeline

The computational procedures were based on four well-documented convolutional neural network architectures: ResNet-50, ConvNeXt-Small, EfficientNet-B3 and DenseNet-121. All networks

were initialised using the *Transfer Learning* technique with pre-trained weights. During the 10-epoch training phase, the model weights were updated by optimising the *Cross-entropy Loss function*. To improve the networks' ability to generalise and prevent overfitting, training was supported by regularisation techniques such as *Early Stop*, which stops training if results do not improve after a certain number of epochs (three in our case), and *Dropout*, which randomly deactivates neurons within the network with $p = 0,5$.

2.4. Uncertainty Management

To overcome the limitations of traditional neural networks, which tend to produce a classification even in the presence of visual artefacts or ambiguous patterns, the methodology incorporates a safety module that calculates a threshold based on the statistical tolerance of the confidence distribution. This threshold is determined by the following formula:

$$Threshold = \mu - K \times \sigma$$

Where μ and σ are calculated from the maximum output value of the *Softmax* function for each image. Predictions with a confidence score lower than the value calculated using this formula are excluded from the three main diagnoses and grouped into a rejection category labelled 'UNKNOWN'. The analyses in this study were conducted with $K = 0,5$.

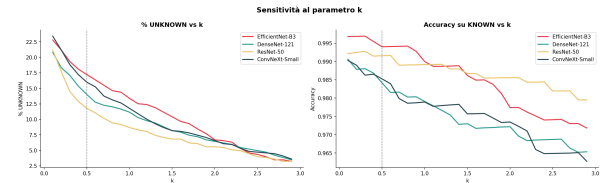


Figure 2: Analysis of K trends

The Figure (2) illustrates the proportion of X-rays rejected by the model as the K parameter varies. The curve slopes downwards. When K is very low, the model is extremely strict, resulting in increased accuracy on the KNOWN set. As the parameter increases, however, the model becomes increasingly permissive, resulting in decreased accuracy on the KNOWN set. The threshold $K = 0.5$ represents the "Sweet Spot", that is, the compromise that balances the two requirements: diagnoses must be correct (high accuracy), but not every ambiguous im-

age can be sent to the doctor (manageable % UNKNOWN).

3. Analysis of Result

As we can see in Table (1), the tests conducted show that all architectures achieved very high levels of accuracy, confirming the models' ability to distinguish successfully between the three classes.

Model	Accuracy	Precision	Recall	F1-Score
EfficientNet-B3	0.961	0.961	0.961	0.961
DenseNet-121	0.949	0.949	0.949	0.948
ResNet-50	0.971	0.972	0.971	0.971
ConvNeXt-Small	0.951	0.951	0.950	0.951

Table 1: Model performance comparison

To gain a deeper understanding of how well the models perform, a confusion matrix (see Fig. (3)) was generated for each architecture on the test set. Analysis of this matrix reveals a near-complete separation for the pneumonia class, which has the fewest false negatives. This confirms that the inflammatory patterns of viral pneumonia are clearly distinct from those of SARS-CoV-2 infection.

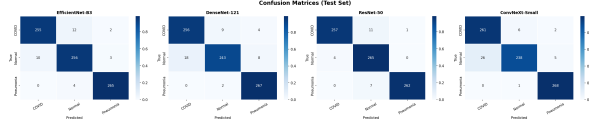


Figure 3: Confusion matrices for each CNN

However, there is some overlap between the 'Covid' and 'Normal' classes. Specifically, a small percentage of cases classified as 'normal' may actually be early-stage infections or images with poor resolution. It is precisely in these cases that the UNKNOWN class is needed to ensure patient safety. Figure (4) shows the model's confidence distribution for each real class and the location of the threshold. Everything to the left of the black line is discarded and classified as UNKNOWN. Note that, unlike the other two classes, there are very few observations to the left of the threshold for the Pneumonia class. This means that the model is almost always certain when it detects pneumonia, discarding very few cases.

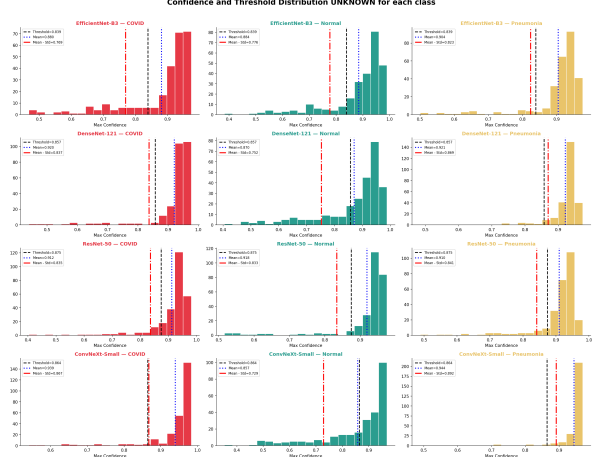


Figure 4: Dynamic threshold for each model

Model	Mean	Std	Threshold
EfficientNet-B3	0.889	0.101	0.838
DenseNet-121	0.903	0.091	0.857
ResNet-50	0.913	0.076	0.874
ConvNeXt-Small	0.913	0.098	0.864

Table 2: Threshold statistics

4. Errors Analysis and XAI

This section analyses the causes and dynamics that cause the model to reject certain images. It explores the relationship between the physical characteristics of the data, the topology of the latent space and visual interpretability. Understanding why an artificial intelligence system fails or is uncertain is fundamental to its clinical applicability.

4.1. UNKNOWN'S Distribution

An analysis of rejection counts reveals varying performance across the architectures. In addition to excelling in its metrics, the ResNet-50 model proved to be the most decisive, rejecting the fewest images overall (95) and balancing rejections evenly across the three real classes (39 for 'Covid', 25 for 'Normal' and 31 for 'Pneumonia'). In contrast, models such as ConvNeXt-Small exhibited significant uncertainty regarding the healthy subject class, rejecting up to 94 real 'normal' images. For this reason, ResNet-50 was chosen as the benchmark for the subsequent error analysis.

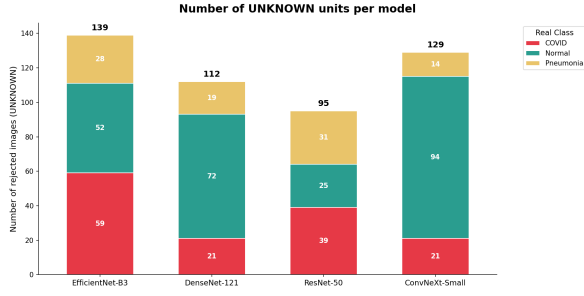


Figure 5: UNKNOWNs for each model

4.2. Analysis of Rejected Images

An analysis of physical features, such as pixel intensity, brightness and contrast, revealed that the distributions of the rejected images are nearly identical to those of the accepted images (see Fig. (6)). This suggests that the model’s uncertainty is not caused by macro-level acquisition errors (e.g. images that are too dark or saturated), but by higher-level factors. For this reason, advanced techniques such as t-SNE and Grad-CAM were employed.

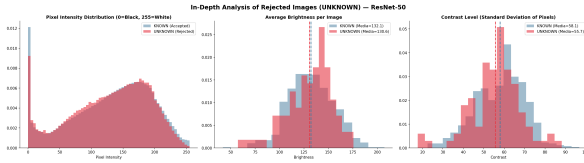


Figure 6: Pixels Distribution of Rejected Images

4.3. t-SNE

t-SNE is an algorithm that shows how the model clusters images prior to decision-making. It reduces the dimensionality of the neural network by transforming each X-ray into a mathematical vector and mapping it onto a 2D plot. This process keeps images that the model considers ‘similar’ close together while separating those that are ‘different’. Looking at Figure fig:tsne, we can see that the blue points (KNOWN) form three distinct clusters: the group in the bottom left dominated by circles (Covid), the group in the top centre dominated by squares (normal) and the isolated group on the right dominated by triangles (pneumonia). This confirms that, during training, the neural network effectively learned to extract valid anatomical features to distinguish between the three pathologies in the vast

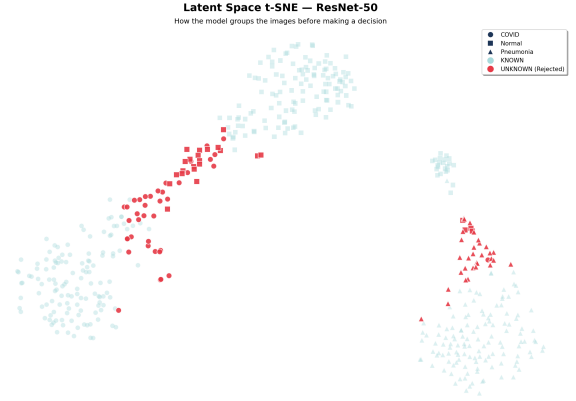


Figure 7: Latent Space of t-SNE

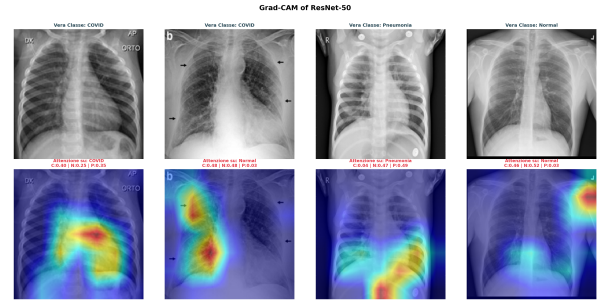


Figure 8: Grad-CAM of Rejected Images

majority of cases. We also note that the red points (UNKNOWN) are positioned exactly on the boundaries and in the transition zones between the main clusters, forming a sort of ‘bridge’. Therefore, the red points are not embedded images, but borderline cases. For example, a paediatric X-ray of a healthy patient may resemble a pulmonary opacity due to the poor quality of the image. The threshold concept was designed precisely to capture these borderline cases and avoid misleading results.

4.4. Grad-CAM

Gradient-weighted Class Activation Mapping (Grad-CAM) is a technique belonging to the field of explainable artificial intelligence (XAI) that generates heatmaps to visualise the regions of an image that are important to the decisions made by CNN models. It uses the gradients of the final convolutional layer to highlight where the network focuses its attention.

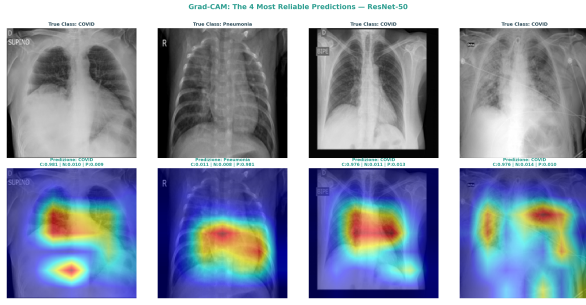


Figure 9: Grad-CAM of Accepted Images

A comparison of Grad-CAM activation maps for ambiguous cases (see Fig. (8)) and reliable cases (see Fig. (9)) reveals the clinical origin of the uncertainty.

Reliable predictions: When confidence is high, the model focuses on specific parenchymal areas and correctly identifies typical pulmonary consolidations.

Ambiguous cases: In rejected images, the network’s attention appears concentrated on non-pathological features, such as the cardiac shadow in paediatric patients or bony artefacts.

5. Conclusion

The initial goal of this study was to assess whether artificial intelligence could classify lung diseases and recognise its own limitations. The results exceeded expectations: modern deep learning architectures were highly effective at feature extraction, achieving very high levels of accuracy. While these peak performance results may have been partly influenced by the relatively small size of our dataset, the real accomplishment lies in error handling. Introducing a dynamic threshold combined with topological space analysis (t-SNE) and visual interpretability (Grad-CAM) showed that the model rejects cases that are anatomically or qualitatively borderline. This approach provides radiologists with a transparent, collaborative diagnostic tool that actively supports them in ambiguous cases rather than replacing them, thereby enhancing the system’s overall clinical safety.

References

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